HDOC Day-52

Previous Greater Element Using Pointers and Nested Loops in C

Problem Statement

Read a variable-length integer array and pass its address to previous_greater(). For each element, display the nearest strictly greater element to its left, or -1 if none exists. Use nested loops and print comma-separated results.

Algorithm

1. Start and read the size of the array; reject a non-positive or excessively large size.
2. Declare a variable-length integer array arr[size] and read all its elements.
3. Pass the address of the first element and the size to previous_greater().
4. For each index i, initialize answer to -1.
5. Using a nested loop, search from j = i - 1 down to 0 for the nearest element strictly greater than arr[i].
6. If found, store arr[j] in answer and stop the inner loop; otherwise retain -1.
7. Print each answer in array order, separated by commas; stop.

Test Case Table

Test Case	Size	Array / Input	Expected Output
1	4	4 5 2 25	-1, -1, 5, -1
2	4	13 7 6 12	-1, 13, 7, 13
3	4	1 3 2 4	-1, -1, 3, -1
4	5	1 2 3 4 5	-1, -1, -1, -1, -1
5	5	5 4 3 2 1	-1, 5, 4, 3, 2
6	4	5 5 3 3	-1, -1, 5, 5
7	1	9	-1
8	5	-2 -5 -1 -3 -4	-1, -2, -1, -1, -3
9	0	(no elements)	Invalid size. Enter an integer from 1 to 10000.
10	3	1 2 x	Invalid array element.

C Program

```
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <stdint.h>

void previous_greater(const int *arr, int size)
{
    int i, j, answer;
    printf("Previous greater elements: ");
    for (i = 0; i < size; i++)
    {
        answer = -1;
        for (j = i - 1; j >= 0; j--)
        {
            if (*(arr + j) > *(arr + i))
            {
```

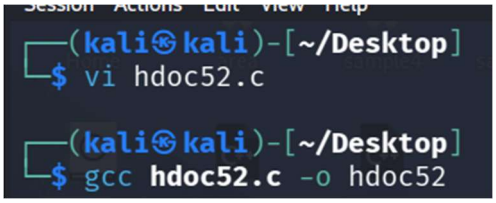
```
                answer = *(arr + j);
                break;
            }
        }
        if (i > 0) printf(", ");
        printf("%d", answer);
    }
    printf("\n");
}

int main(void)
{
    int size, i;
    printf("Enter array size: ");
    if (scanf("%d", &size) != 1 || size <= 0 || size > 10000)
    {
        printf("Invalid size. Enter an integer from 1 to 10000.\n");
        return 1;
    }
    int arr[size];
    printf("Enter %d array elements:\n", size);
    for (i = 0; i < size; i++)
    {
        if (scanf("%d", arr + i) != 1)
        {
            printf("Invalid array element.\n");
            return 1;
        }
    }
    previous_greater(arr, size);
    return 0;
}
```

Compilation and Execution Commands

```
gcc -std=c99 hdoc52.c -o hdoc52
./hdoc52
```

Compilation and Execution



Test Case 1

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 4
Enter 4 array elements:
4 5 2 25
Previous greater elements: -1, -1, 5, -1
```

Test Case 2

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 4
Enter 4 array elements:
13 7 6 12
Previous greater elements: -1, 13, 7, 13
```

Test Case 3

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 4
Enter 4 array elements:
1 3 2 4
Previous greater elements: -1, -1, 3, -1
```

Test Case 4

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 5
Enter 5 array elements:
1 2 3 4 5
Previous greater elements: -1, -1, -1, -1, -1
```

Test Case 5

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 5
Enter 5 array elements:
5 4 3 2 1
Previous greater elements: -1, 5, 4, 3, 2
```

Test Case 6

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 4
Enter 4 array elements:
5 5 3 3
Previous greater elements: -1, -1, 5, 5
```

Test Case 7

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 1
Enter 1 array elements:
9
Previous greater elements: -1
```

Test Case 8

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 5
Enter 5 array elements:
-2 -5 -1 -3 -4
Previous greater elements: -1, -2, -1, -1, -3
```

Test Case 9

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 0
Invalid size. Enter an integer from 1 to 10000.
```

Test Case 10

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc52
Enter array size: 3
Enter 3 array elements:
1 2 x
Invalid array element.
```